Aust Zahle

Faculty of engineering

ICT

Fadi al kadi

ID: 22330019

Part a

```
FLARE-VM Mon 03/09/2026  2:38:35.02
C:\Users\FADI\Desktop>hostname && whoami && getmac
ID-22330019
id-22330019\fadi

Physical Address    Transport Name
=================== ==========================================================
08-00-27-60-0C-C4   \Device\Tcpip_{BD07F2C5-73E8-427D-B615-C1F992374D9E}
N/A                 Media disconnected
00-FF-2F-9C-DB-81   Media disconnected
N/A                 Media disconnected
```

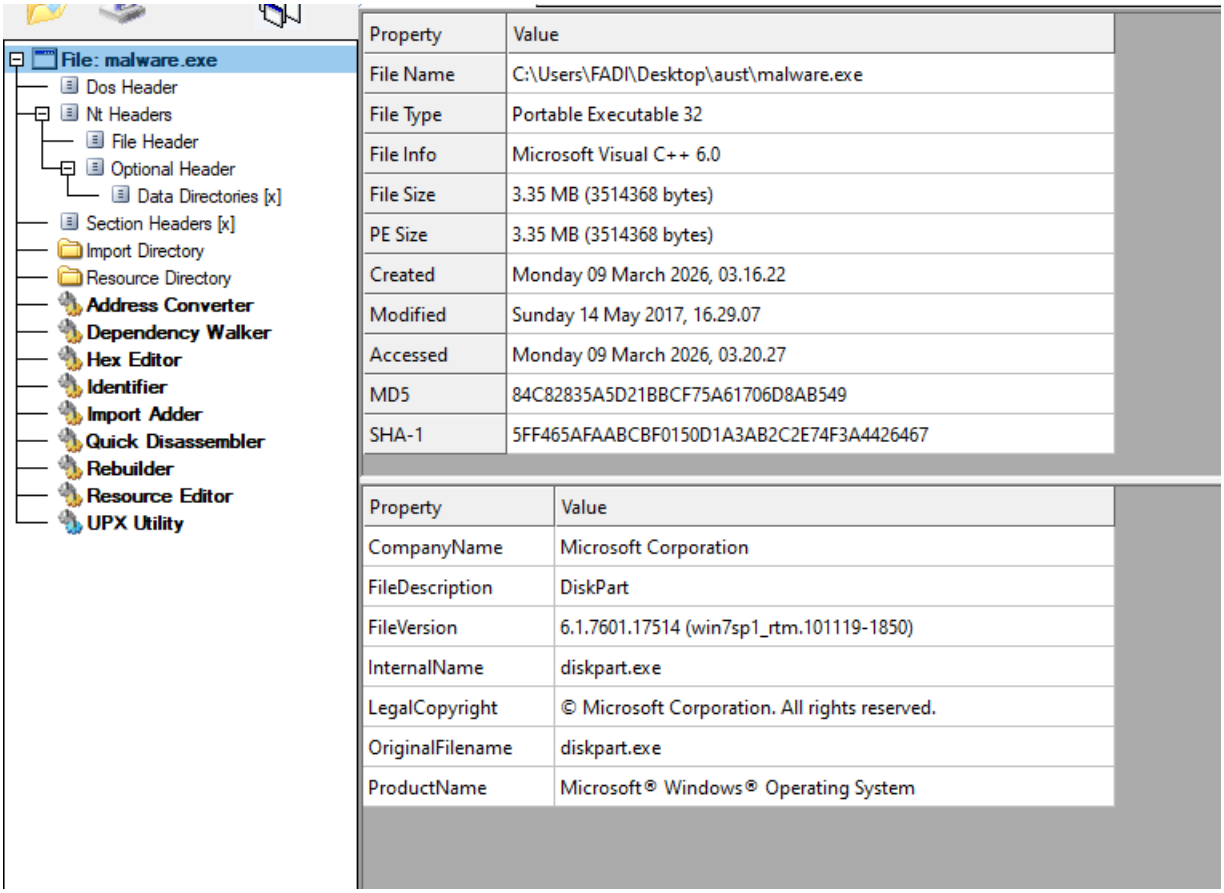
```
C:\Users\FADI\Desktop\aust>dir
 Volume in drive C has no label.
 Volume Serial Number is B08B-B949

 Directory of C:\Users\FADI\Desktop\aust

03/09/2026  02:59 AM    <DIR>          .
03/09/2026  02:59 AM    <DIR>          ..
03/09/2026  02:59 AM           200,319 Asssignment-1-instructions.docx
03/09/2026  02:59 AM         4,165,391 companion_guide.pdf
03/09/2026  02:59 AM         3,481,026 sample.zip
               3 File(s)      7,846,736 bytes
               2 Dir(s)  56,037,191,680 bytes free
```

```
FLARE-VM Mon 03/09/2026  3:09:59.78
C:\Users\FADI\Desktop\aust>certutil -hashfile malware.exe SHA256
SHA256 hash of malware.exe:
ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa
CertUtil: -hashfile command completed successfully.

FLARE-VM Mon 03/09/2026  3:16:43.10
C:\Users\FADI\Desktop\aust>certutil -hashfile malware.exe MD5
MD5 hash of malware.exe:
84c82835a5d21bbcf75a61706d8ab549
CertUtil: -hashfile command completed successfully.
```



Property	Value
File Name	C:\Users\FADI\Desktop\aust\malware.exe
File Type	Portable Executable 32
File Info	Microsoft Visual C++ 6.0
File Size	3.35 MB (3514368 bytes)
PE Size	3.35 MB (3514368 bytes)
Created	Monday 09 March 2026, 03.16.22
Modified	Sunday 14 May 2017, 16.29.07
Accessed	Monday 09 March 2026, 03.20.27
MD5	84C82835A5D21BBCF75A61706D8AB549
SHA-1	5FF465AFAABCBF0150D1A3AB2C2E74F3A4426467

Property	Value
CompanyName	Microsoft Corporation
FileDescription	DiskPart
FileVersion	6.1.7601.17514 (win7sp1_rtm.101119-1850)
InternalName	diskpart.exe
LegalCopyright	© Microsoft Corporation. All rights reserved.
OriginalFilename	diskpart.exe
ProductName	Microsoft® Windows® Operating System

String Value	Category	Significance
WanaCrypt0r	Ransomware Identifier	Name associated with WannaCry ransomware
WANACRY!	Ransomware Identifier	Direct reference to WannaCry malware
msg/m_english.wnry	File Path	Language ransom note file used by ransomware
cmd.exe /c "%s"	Command	Indicates malware executes system commands
icacls . /grant Everyone:F /T /C /Q	Command	Changes file permissions to allow full access
CryptAcquireContextA	Encryption API	Windows crypto function used for encryption
WanaCrypt0r@2017	Encryption String	Encryption component used by WannaCry
.wnry	File Extension	Ransomware file extension for encrypted files

```
FLARE-VM Mon 03/09/2026  3:27:54.82
C:\Users\FADI\Desktop\aust>floss.exe malware.exe > floss_output.txt
INFO: floss: extracting static strings
finding decoding function features: 100%|▉| 137/137 [00:00<00:00, 1082.68 functions/s, skipped 6 library functions (4%)
INFO: floss.stackstrings: extracting stackstrings from 102 functions
INFO: floss.results: oftware\
extracting stackstrings: 100%|                                        | 102/102 [00:00<00:00, 1037.59 functions/s]
INFO: floss.tightstrings: extracting tightstrings from 13 functions...
extracting tightstrings from function 0x406880: 100%|                 | 13/13 [00:02<00:00,  4.92 functions/s]
INFO: floss.string_decoder: decoding strings
emulating function 0x403a28 (call 2/2): 100%|                         | 22/22 [00:01<00:00, 12.54 functions/s]
INFO: floss: finished execution after 20.36 seconds
INFO: floss: rendering results
```

String Value	Category	Significance / Notes
diskpart.exe	Masquerading	The malware pretends to be a legitimate Windows tool to avoid suspicion
.docx .xlsx .pptx .pdf .txt	File Extensions	Indicates the malware targets user documents
.sqlite3 .db .pst	File Extensions	Suggests database and email files are also targeted
CryptAcquireContext	Encryption API	Windows cryptography function used to generate encryption keys
CryptGenKey	Encryption API	Indicates encryption capability for ransomware behavior
CreateServiceA	Persistence	Suggests malware may create a service to remain on the system
OpenSCManagerA	Service Control	Shows ability to interact with Windows services
Software\	Registry Path	Indicates possible registry modifications
cmd.exe /c "%s"	Command Execution	Malware may execute system commands through CMD
GetComputerNameW	System Info	Used to collect information about the infected system

Field	Value
File Name	malware.exe
File Size	3.35 MB (3514368 bytes)
File Type (magic bytes)	Portable Executable 32 (PE32 Windows executable)
MD5 Hash	84C82835A5D21BBCF75A61706D8AB549
SHA-256 Hash	ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa
PE Architecture	x86 (32-bit)
Compile Timestamp	Sunday 14 May 2017 16:29:07
Number of Sections	5 (.text, .rdata, .data, .rsrc, .reloc)
Names of Imported DLL Libraries	KERNEL32.dll, USER32.dll, ADVAPI32.dll, SHELL32.dll, OLEAUT32.dll, WS2_32.dll
Threat Category	Ransomware

String Value	Category	Significance / Notes
.docx .xlsx .pptx .pdf .txt	File Extensions	Indicates the malware targets common user document files
.sqlite3 .pst .edb	File Extensions	Shows the malware targets databases and email storage files
CryptAcquireContext	Encryption API	Used to initialize cryptographic operations
CryptGenKey	Encryption API	Generates encryption keys used to encrypt files
cmd.exe /c "%s"	Command	Indicates the malware can execute system commands
CreateServiceA	Persistence	Suggests malware may create a service to run automatically
OpenSCManagerA	Service Control	Allows the malware to manage Windows services
Software\	Registry Path	Indicates possible registry modifications
GetComputerNameW	System Information	Used to gather information about the infected machine
diskpart.exe	Masquerading	Malware pretends to be a legitimate Windows system utility

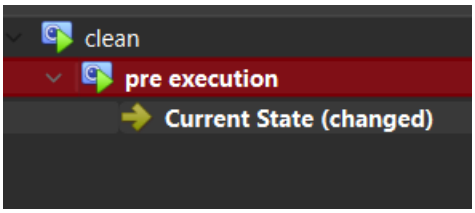
The malware appears to be **ransomware**. Evidence supporting this includes the presence of multiple targeted file extensions such as .docx, .xlsx, .pptx, .pdf, and .txt, which are commonly encrypted by ransomware. Additionally, cryptographic functions such as CryptAcquireContext and CryptGenKey indicate that the program performs encryption operations, which is a typical behavior of ransomware used to lock victim files and demand payment.

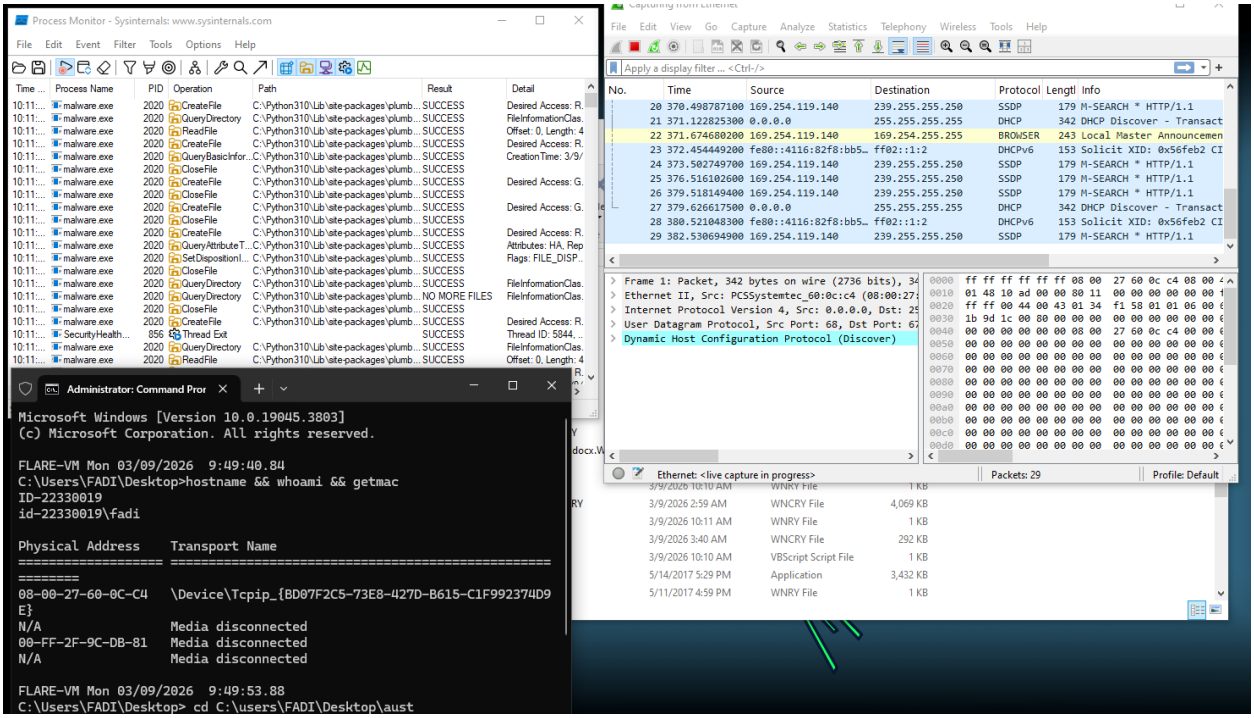
FLOSS can extract **obfuscated and dynamically generated strings** that standard string extraction tools cannot detect. The strings tool only extracts static readable text embedded directly in the binary file. Malware developers often hide important strings using encoding or runtime construction techniques. FLOSS analyzes the program logic and reconstructs these hidden strings, revealing indicators such as commands, URLs, or registry paths that would otherwise remain hidden.

The string cmd.exe /c "%s" indicates that the malware can execute commands through the Windows command interpreter. This suggests that the malware may run system commands, execute scripts, or launch additional payloads during the infection process.

Part b

Snapshots:





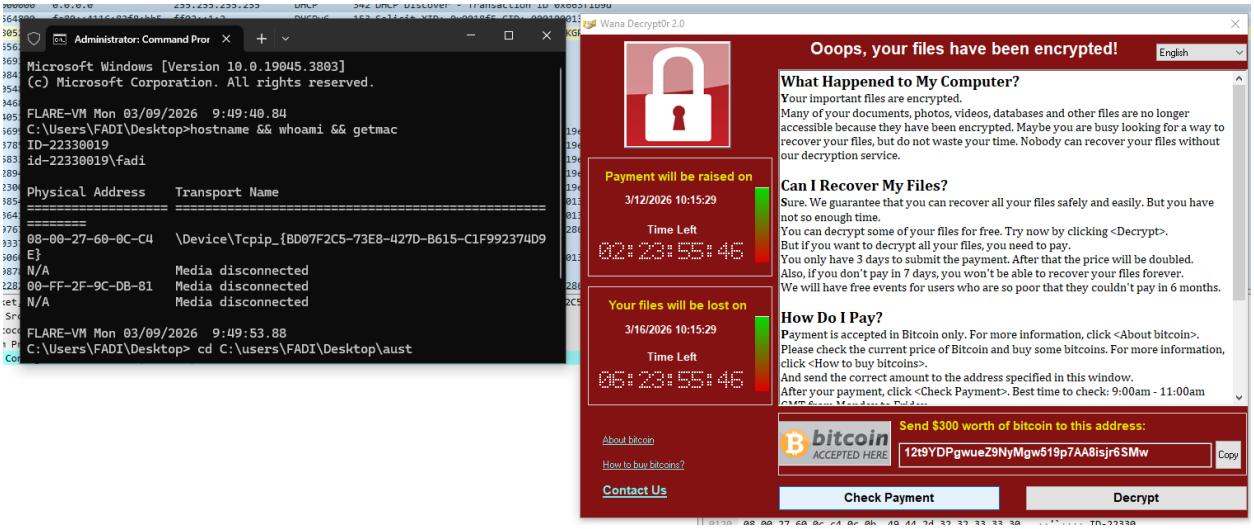
Observation	Explanation
Multiple CreateFile operations	The malware opens and creates files during execution
Repeated QueryDirectory operations	Indicates the malware is scanning directories for files
ReadFile operations	Suggests the malware reads file contents before modifying them
Access to multiple directories	Indicates enumeration of files on the system
High number of file operations	Typical behavior of ransomware searching for files to encrypt

Registry Operation Explanation

RegCreateKey	Malware may create registry keys for persistence
RegSetValue	Used to modify system configuration
RegQueryValue	Used to check system settings or installed software

Observation	**Explanation**
DHCP Discover packets	VM attempting to obtain an IP address
SSDP M-SEARCH requests	Device discovery protocol used in local networks
Workgroup announcements	Windows network discovery behavior
No external HTTP or DNS communication	Network isolation prevented internet access

During dynamic analysis, the malware generated a large number of file system operations, including file creation, reading, and directory enumeration. This behavior suggests that the malware scans the system for files to process. Network analysis using Wireshark showed only local broadcast traffic such as DHCP and SSDP requests. No external communication with command-and-control servers was observed due to the isolated network configuration used during analysis.



The restoration of the vm through snapshot:

Part c:

How did static analysis help you understand the malware?

Static analysis helped identify important characteristics of the malware **without executing the file**. By analyzing the executable with tools such as CFF Explorer, strings, and FLOSS, it was possible to observe the file structure, imported libraries, and embedded strings. The presence of cryptographic functions like CryptAcquireContext and CryptGenKey, along with targeted file extensions such as .docx, .xlsx, .pdf, and .txt, suggested that the malware performs file encryption. Static analysis also revealed that the executable attempts to masquerade as a legitimate Windows program (diskpart.exe), which is a common technique used by malware to avoid suspicion.

What behaviors were observed during dynamic analysis?

During dynamic analysis, the malware generated a large number of file system operations observed in Process Monitor. These operations included CreateFile, ReadFile, QueryDirectory, and CloseFile, indicating that the malware scans directories and accesses files on the system. This behavior is consistent with ransomware that searches for user documents to encrypt. Network monitoring using Wireshark showed only local broadcast traffic such as DHCP and SSDP packets. No external connections were observed because the virtual machine was configured with an isolated internal network, preventing communication with external servers.

Why is it important to analyze malware in an isolated environment?

Analyzing malware in an isolated environment is essential to prevent the malware from spreading or causing damage to real systems. A virtual machine with no internet connectivity ensures that the malware cannot communicate with command-and-control servers or infect other devices on the network. Isolation also allows the analyst to safely observe malicious behavior while protecting the host system and surrounding infrastructure.

What limitations did you encounter during the analysis?

One limitation encountered during the analysis was the lack of external network communication due to the isolated network configuration. While this is important for safety, it also prevents observing how the malware interacts with command-and-control servers or downloads additional payloads. Additionally, some malware strings may be obfuscated or dynamically generated, which makes them difficult to detect using simple static analysis tools.

This analysis demonstrated how both static and dynamic techniques can be used to investigate malicious software. Static analysis allowed the identification of suspicious indicators such as encryption functions and targeted file extensions, while dynamic analysis revealed the runtime behavior of the malware, including file system activity and system interactions. Conducting the analysis in an isolated virtual environment ensured that the malware could be studied safely without risking infection of the host system.